ARCHIT AGRAWAL

San Francisco Bay Area | 623-312-0435 | architagrawal000@gmail.com | LinkedIn | GitHub | Portfolio

PROFESSIONAL EXPERIENCE

EdPlus, Arizona State University **06/2026 - Present**
AI Software Engineer *Phoenix, AZ*

- Own architecture and delivery of an AI analytics platform on AWS Bedrock and Step Functions that turns any structured dataset into cited, code-computed answers.
- Built the platform's web application with a Nuxt frontend and NestJS backend, sharing TypeScript contracts across 70 API routes.
- Reduced AWS pipeline runtime from 500 to 85 seconds by replacing two managed-agent stages with direct tool calls.
- Cut incorrect answers 85% in an evaluation against 164 published benchmark figures by adding answer verification before display.
- Validated 25 statistical tests and six causal estimators against SciPy and statsmodels; scored 68% on a 100-question StatQA sample, against GPT-4o's best reported 64.83%.
- Improved accuracy from 60% to 93% on the hardest benchmark questions by consolidating column definitions in a shared semantic layer.
- Quadrupled the headline findings automated reports recover, 13% to 52–59% on held-out datasets, by letting an LLM choose analyses while code computes every number.
- Reached 100% join integrity on repeat runs with schema-constrained LLM output and per-row IDs; the prior pipeline silently mislabeled 1 in 8 rows.
- Showed model choice moves labeling F1 33x more than pipeline design (0.230 vs 0.007), redirecting the team toward reliability.

MyStage Music Inc. **03/2025 - 06/2026**
Founding AI/ML Engineer *Remote*

- Consolidated a 14-Cloud-Function backend into a checkpointed Python/LangGraph workflow on GCP Cloud Run, reducing production service dependencies 64%.
- Built FastAPI and Algolia search with sub-50 ms latency, real-time Firestore synchronization, 200+ pytest tests and Logfire tracing.
- Scaled ingestion to 70,000+ records daily from 650+ venues using Playwright, transactional locking and entity resolution.
- Implemented checkpointed pause-and-resume so records missing context wait for additional information without failing the ingestion batch.
- Made ingestion retries safe with deterministic task IDs and atomic create-if-absent writes, preventing duplicate jobs from overwriting in-flight work.

EdPlus, Arizona State University **09/2023 - 05/2025**
Software Engineer *Phoenix, AZ*

- Built a multi-tenant course assistant from React UI to retrieval services, used by 1,000+ faculty creating courses for 60,000+ students.
- Established Prompt Flow release checks for groundedness, relevance and coherence, evaluating prompt changes against measured results before deployment.
- Implemented tenant isolation in retrieval services and configuration-based onboarding for additional colleges.
- Added hybrid keyword and vector retrieval to resolve exact course codes alongside semantic document search.
- Reduced transcript-analysis time from four hours to 15 minutes using a Neo4j knowledge graph and validated Cypher queries.

Arizona State University **08/2024 - 05/2025**
Student Researcher *Phoenix, AZ*

- Built a PyTorch and Diffusers benchmark for text-to-video models that scores physical plausibility separately from visual quality, holding seed, resolution and sampler fixed.
- Found visual quality barely predicts physical plausibility: the top model by human preference ranked last on causal consistency.

- Designed a frame-level rubric for rigid-body motion, gravity, momentum and occlusion, and measured inter-rater agreement before trusting any aggregate score.

Knowledge Exchange for Resilience, Arizona State University **06/2024 - 08/2024**
Data Research Aide *Phoenix, AZ*

- Reduced p95 retrieval latency 60% at 95% recall using hybrid FAISS and Neo4j search.
- Built a FastAPI and pgvector collaborator recommender achieving 85% top-k precision in evaluation.
- Cut API latency from 198 to 20 ms with Redis caching and SQL rewrites, verified under simulated peak load with Locust.

Zeus Learning **01/2022 - 07/2023**
Software Engineer *Mumbai, India*

- Reduced deployment time 70% and production incidents 40% through containerized CI/CD and SonarQube quality gates.
- Migrated a C#/.NET monolith to Dockerized Kubernetes microservices, reducing infrastructure cost 20% and resource footprint 35%.
- Built a Redis-backed desk-booking service used across 300+ offices at Fortune 500 companies.

PROJECTS

SRP Electric MCP Server | GitHub **12/2025 - 01/2026**

- Built a Python/FastMCP integration exposing seven utility-account tools for billing, usage and outage queries, with an interactive setup wizard.
- Reverse-engineered the portal's login flow and API calls, since the utility publishes no API documentation.

MCP GitHub PR Review Agent **07/2025 - 08/2025**

- Built a TypeScript PR-review integration combining code diffs, tickets and CI results into schema-validated inline findings.
- Made review comments idempotent across pushes and synchronized ticket context with Asana.

AiJockey – AI DJ Pipeline | GitHub **2025 - Present**

- Built a PyTorch/FastAPI audio pipeline with stem separation, LLM transition planning and 25+ DSP transitions; ported inference to AMD MI300X.
- Trained a MERT-95M reward head to score audio segments across four aesthetic dimensions for candidate selection.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C#
AI & LLM: AWS Bedrock, LangGraph, RAG, LLM evaluation, Prompt Flow, MCP, PyTorch
Retrieval & Data: PostgreSQL, pgvector, FAISS, Neo4j, Redis, Algolia, Firestore
Backend & Web: FastAPI, NestJS, Node.js, .NET, React, Nuxt
Cloud & Quality: AWS (Step Functions, Lambda, S3, CDK), GCP Cloud Run, Docker, Kubernetes, CI/CD, pytest

EDUCATION

Arizona State University (ASU) **08/2023 - 05/2025**
M.S., Computer Science | GPA: 4.0 *Tempe, AZ*

Dhirubhai Ambani Institute of Information and Communication Technology **08/2018 - 05/2022**
B.Tech., Information and Communication Technology *Gandhinagar, India*